Question 1

(a) True

(b) True

(c) False

(d) False

Question 2

We prove this true using the subgroup test:
First note that since $H \leq G$, $1_G \in H$. Since φ is homomorphic, $\varphi(1_G) = 1_{G'} \in \varphi(H)$. Now let $h_1, h_2 \in \varphi(H)$. Then there exists some $g_1, g_2 \in H$ such that $\varphi(g_1) = h_1$ and $\varphi(g_2) = h_2$. But then g_2^{-1} exists (since H is a group) and so since φ is a homomorphism, we deduce

$$\varphi(g_1 g_2^{-1}) = \varphi(g_1)\varphi(g_2^{-1}) = \varphi(g_1)\varphi(g_2)^{-1} = h_1 h_2^{-1} \Rightarrow h_1 h_2^{-1} \in \varphi(H)$$

making $\varphi(H)$ a subgroup of G'

Question 3

Consider $\varphi : (\mathbb{R}, +) \to (\{0\}, +)$ defined by $\varphi(x) = 0$. This is a homomorphism, since for $x, y \in \mathbb{R}$, $\varphi(x + y) = 0 = 0 + 0 = \varphi(x) + \varphi(y)$. But, for example, $H = \{2\}$ we get $\varphi(H) = \{0\} \leq \{0\}$ but H is not a subgroup of $\mathbb{R}$ as $0 \notin H$. Thus this does not hold